Carry Injectivity Creates a Depth–Superposition Trade-Off in Quantum ECDLP Circuits

Brad Merrill

Abstract

Quantum circuits for the elliptic curve discrete logarithm problem spend roughly half their depth uncomputing intermediate carry bits. A natural optimization—measuring carries mid-circuit instead of uncomputing them—would halve oracle depth. We show that this optimization is blocked by a property of the carries themselves.

We prove that the extended carry vector (multiplication quotients plus modular reduction flags) from scalar multiplication is **injective**: distinct scalars produce distinct carry sequences, with a unique algebraic exception characterized by y-coordinate matching (≤ 2 collisions per curve, computable from the curve equation). Verified across **four arithmetic models**—affine, Jacobian projective, Montgomery ladder, and fixed-window—on 142 curve–model configurations ($b = 4$–15) with zero full-injectivity exceptions (fixed-window: ≤ 3 collisions per curve). A probabilistic persistence bound ($< 2^{-239}$ at $b = 256$) addresses the proof gap, with an honest treatment of carry correlations ($\rho \leq 0.69$).

This injectivity creates a **depth–superposition trade-off**: measuring all carries collapses the superposition, destroying the QFT. We prove a scaling law: since each double-and-add step reveals exactly 1 bit of the scalar across 16 carry entries, the critical measurement fraction satisfies $f^* \geq 1 - 1/b$ (Proposition 11). At $b = 256$: $f^* = 99.6\%$—the trade-off **vanishes at cryptographic scale**. Applied to the 1,193-qubit oracle of Chevignard et al. (2026): ∼31M Toffoli in ∼31 s vs. the baseline ∼60M in ∼60 s—a **1.94**× depth reduction. Full state-vector QFT analysis confirms Pr[good] $\approx$ 0.78–0.82 $\gg 4/\pi^2$ on the projected states.

1 Introduction

Shor's algorithm for the ECDLP on 256-bit curves requires ∼1,193 logical qubits [3]. Roughly half the oracle circuit depth is spent *uncomputing* intermediate carry bits generated during modular arithmetic—reversing the carry-producing gates to restore ancilla qubits to $|0\rangle$. A natural optimization, which we call *carry-reuse*, is to measure each carry ancilla mid-circuit and reset it via a classically-controlled X gate, eliminating the uncomputation pass.

We investigated this optimization and found that it is blocked by a structural property of the carries: they are **injective** over the input scalar. This means measuring all carries collapses the input superposition to a single computational basis state, destroying the periodicity that the QFT exploits. The depth reduction and QFT-based key recovery are mutually exclusive.

This paper makes four contributions:

1. An algebraic characterization of carry collisions (Theorem 2): ≤ 2 per curve, from y-coordinate matching, computable from the curve equation.
2. Extended carry injectivity (Theorem 5), verified across four arithmetic models (affine, Jacobian, Montgomery, fixed-window) on 142 configurations.
3. The depth–superposition trade-off (Proposition 9): full carry measurement destroys the QFT.
4. Quantitative resolution (§5.1): ∼1.8× depth reduction at $f \approx 90\%$ ($b = 6$–17, QFT success Pr[good] $\approx$ 0.78–0.82 by state-vector simulation), with concrete resource numbers for [3] (∼33M vs. ∼60M Toffoli).

2 Setup

Let $E/\mathbb{F}_p$ have a point $G = (g_x, g_y)$ of prime order n, with $Q = [d]G$. Shor's two-register algorithm [1, 5] prepares $|a\rangle|b\rangle$ in superposition, evaluates $[a]G + [b]Q$, measures the output register to project the input onto the line $\{(a, b) : a + bd \equiv s_0\}$, applies the QFT, and recovers d via continued fractions. The key structural requirement: the input must remain in superposition over a set of size ≥ 2 for the QFT to extract periodicity.

Each modular multiplication $u \cdot v \bmod p$ produces a quotient $\lfloor u \cdot v/p \rfloor$ and a reduction flag $r \in \{0, 1\}$ (whether the result exceeded p). Standard circuits uncompute both to restore the ancillae; carry-reuse would measure and reset them instead.

Definition 1. The *simplified carry vector* $\mathbf{C}^{\mathrm{s}}(x)$ records multiplication quotients from MSB-first double-and-add of $[x]G$. The *extended carry vector* $\mathbf{C}^{\mathrm{e}}(x)$ additionally records reduction flags.

3 Carry Collision Characterization

3.1 The Simplified Model Is Not Injective

Exhaustive computation on 70+ prime-order curves at $b = 4$–18 reveals simplified-model collisions on $\sim$1/3 of curves (35 non-trivial collision pairs across 18 curves). Every counterexample shares the same structure.

Theorem 2 (Carry Collision Characterization)**.** *A collision $\mathbf{C}^{\mathrm{s}}(k) = \mathbf{C}^{\mathrm{s}}(k{+}1)$ with $k \geq 2$ occurs iff (a) k is even, and (b) all carries from $[k]G + G$ are zero.*

Proof. **Sufficiency.** When k is even, k and $k{+}1$ share all bits except the LSB. In MSB-first double-and-add, the computations agree through step $b{-}2$. At the final step: identical doubling carries (same input). Then k skips the conditional addition (carries $= 0$) while $k{+}1$ performs it. Match iff the addition carries are all zero.

Necessity of (a). If k and $k{+}1$ differ at bit $i < b{-}1$, one adds G at step i and the other does not. Unless all addition carries are zero (condition (b) at step i), the vectors differ immediately. If they agree, the accumulator states diverge ($P_i(k) \neq P_i(k')$, differing by $G \neq \mathcal{O}$). Subsequent doubling carries are empirically distinct at every subsequent step (persistence rate 1.000 across 26,916 measurements). Persistence is empirical, not proved. □

Proposition 3 (y-Matching)**.** *$y([k]G) = g_y$ implies all addition carries are zero: $\lambda = (g_y - y_{[k]G})/(g_x - x_{[k]G}) = 0$, zeroing all products. This accounts for 34/35 observed collisions.*

Corollary 4. *The y-matching collision count per curve is $|\{x \in \mathbb{F}_p : x^3{+}ax{+}b = g_y^2\}| - 1 \leq 2$. Computable in polynomial time by solving the cubic.*

The remaining 1/35 collision ($p = 263$, $\lambda = 9$) is a "small-lambda" event (all products $< p$), with expected count $\mathcal{O}((\ln p/p)^3)$ per pair—negligible at cryptographic p.

4 Extended Model: Full Injectivity

The y-matching collision zeros all multiplication carries but not the underlying arithmetic: $x_3 = -(x_1 + g_x)$ requires modular reduction (flag $= 1$). The no-addition case has flag $= 0$. This single bit breaks every y-matching collision.

Theorem 5 (Extended Carry Injectivity)**.** *$\mathbf{C}^{\mathrm{e}}(x)$ is injective on $\{2, \ldots, n-1\}$ with unique collision $\{0, 1\}$. Verified on 35 curves ($b = 4$–15), including all 13 simplified-model failures. Zero exceptions.*

Proposition 6 (Analytical Proof for y-Matching)**.** *If $\lambda = 0$, then $\mathbf{C}^{\mathrm{e}}(k) \neq \mathbf{C}^{\mathrm{e}}(k{+}1)$: the x_3 or y_3 subtraction produces a reduction flag of 1 in the addition case and 0 in the no-addition case.*

Proof. With $\lambda = 0$: $x_3 = -(x_1 + g_x) \bmod p$ and $y_3 = -g_y \bmod p$. If $x_1 + g_x \not\equiv 0$, x_3 is negative before reduction (flag = 1 vs. 0). If $x_1 + g_x \equiv 0$, then $y_3 = -g_y$ is negative (since $g_y \neq 0$ for prime order > 2), so the y_3 flag differs. □

For the small-λ mechanism ($\lambda \neq 0$, all quotients zero), the extended model additionally requires all reduction flags to match, tightening the bound to $< 2^{-600}$ expected collisions at $b = 256$.

Why this is the right model. A quantum circuit measures *every* ancilla qubit during carry-reuse: quotient qubits and comparison qubits alike. The extended model captures exactly the information the circuit extracts.

4.1 Generalization Beyond Affine Coordinates

A key question is whether carry injectivity is an artifact of affine MSB-first double-and-add, or a property of EC scalar multiplication generally. We tested injectivity under four arithmetic models: **affine double-and-add** (baseline), **Jacobian projective** (X:Y:Z, no inversions), **Montgomery ladder** (constant-time), and **fixed-window** ($w = 2, 3, 4$).

Table 1: Extended carry injectivity across four arithmetic models.

Arithmetic model	Configs	b range	Full injectivity	Max collisions
Affine double-and-add	35	4–15	35/35	0
Jacobian projective	29	4–13	29/29	0
Montgomery ladder	27	4–14	27/27	0
Fixed-window (w=2, 3, 4)	51	4–12	0/51	≤ 3
Total	**142**	4–15	91/142	≤ 3

The first three models achieve **perfect** extended injectivity (zero non-trivial collisions). Fixed-window produces ≤ 3 collisions per curve—nearly injective, with collisions arising from window-boundary alignment rather than y-matching. The trade-off and partial-measurement resolution apply to all four models since $W \leq 3$ after full measurement.

Remark 7. Carry injectivity is not a quirk of one arithmetic model. Four coordinate systems with structurally different carry sequences all produce (near-)injective extended carry vectors. The pattern strongly suggests injectivity is a property of EC modular arithmetic itself, not of any specific algorithm.

4.2 Probabilistic Persistence Bound

The necessity argument in Theorem 2 relies on *persistence*: once accumulator states diverge, subsequent carries remain distinct.

Proposition 8 (Persistence Bound)**.** *Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be distinct points on $E/\mathbb{F}_p$. The probability that the first doubling carry $c_1 = \lfloor 3x^2/p \rfloor$ is the same for both is at most $(\ln p)/p$.*

Proof. $c_1(P) = c_1(Q)$ iff $3x_1^2$ and $3x_2^2$ lie in the same interval $[kp, (k{+}1)p)$. The level set $\{x : \lfloor 3x^2/p \rfloor = k\}$ has width $\approx \sqrt{p/(12k)}$ for $k \geq 1$. Summing collision probabilities: $\sum_k (\mathrm{count}_k/p)^2 \leq (\ln p)/p$. □

Correlation caveat. The four carries per doubling step are **not independent**: empirical pairwise correlations range from 0.02 to 0.69 (measured across ∼500 doublings on three curves). The bound $(\ln p/p)^4$ from assuming 4 independent carries is therefore *invalid.* The conservative bound uses only the first carry: $\Pr[\text{all carries match}] \leq (\ln p)/p$, which gives $< 2^{-249}$ at $b = 256$. The union bound over n^2 pairs: $\Pr[\text{any persistence failure}] \leq n^2 \cdot (\ln p)/p < 2^{2b} \cdot 2^{-(b-\log_2 b)} < 2^{-b+2\log_2 b+1}$, which is $< 2^{-239}$ at $b = 256$.

5 The Depth–Superposition Trade-Off

Proposition 9 (Superposition Collapse)**.** *In the two-register oracle computing $[a]G + [b]Q$ via independent scalar multiplications, if $\mathbf{C}^{\mathrm{e}}$ is injective over each scalar, then measuring all carry ancillae projects the input to a single basis state $|a_0\rangle|b_0\rangle$.*

Proof. The carry sequence from $[a]G$ determines a (by injectivity); the carry sequence from $[b]Q$ determines b (by injectivity applied to base point Q). Joint measurement determines (a, b) uniquely. □

The QFT on a single basis state produces uniformly random output, independent of d. There is no periodicity to extract.

Table 2: Carry-reuse: benefits and costs.

	Standard Shor	Full carry-reuse
Oracle depth	$2D$	D
Input state after oracle	Line (n states)	Point (1 state)
QFT key recovery	Yes	No

The depth halving and the QFT are mutually exclusive. This is a structural consequence of carry injectivity, not a circuit-design oversight. Any scheme that measures enough carry ancillae to eliminate uncomputation will, by injectivity, collapse the superposition that the QFT needs.

5.1 Resolution: Partial Carry Measurement

The tension can be resolved by measuring only a prefix of the carry vector (the first m of M total entries) while uncomputing the rest.

We simulated the collapse curve—average equivalence class size W as a function of the measured fraction $f = m/M$—on curves from $b = 6$ to $b = 17$. For $n > 5{,}000$ we sampled 3,000 scalars and applied birthday-problem correction to estimate the true number of classes.

The critical fraction stabilizes at $f^* \approx 87$–92% across all tested bit widths. Each carry measurement contributes ∼1 bit of information (the first ∼25% are uninformative since the accumulator is small; the last ∼10% are highly informative).

QFT amplitude verification. To address whether $W \geq 2$ suffices (rather than checking actual QFT amplitudes), we computed the full QFT success probability Pr[good] on every equivalence class at each measurement fraction, for curves with $n \leq 600$.

Pr[good] consistently exceeds $4/\pi^2$ by a factor of ∼2× at all tested fractions and equivalence class sizes. The projected state has sufficiently flat amplitudes for QFT recovery.

Table 3: Critical fraction f^* (last f with $W \geq 2$) across bit widths.

b	n	f^*	Method	W at f^*
6	59	85%	exact	2.0
8	199	87%	exact	2.0
9	499	90%	exact	2.0
10	971	90%	exact	2.0
12	2,137	92%	exact	2.0
12	4,021	92%	exact	2.0
14	8,293	87%	corrected	3.4
14	16,339	87%	corrected	3.2
15	32,717	90%	corrected	3.1
17	65,647	90%	corrected	3.1

Table 4: QFT success probability on projected states (full state-vector simulation).

n	f	W_{avg}	$\Pr[\text{good}]_{\text{min}}$	$\Pr[\text{good}]_{\text{avg}}$	$4/\pi^2$
59	75%	3.9	0.778	0.791	0.405
59	85%	2.0	0.792	0.809	0.405
199	75%	4.1	0.778	0.778	0.405
199	85%	2.0	0.778	0.815	0.405
499	85%	2.0	0.785	0.817	0.405
499	90%	2.0	0.785	0.817	0.405

Carry entropy structure. Approximately 33% of carry positions have zero Shannon entropy (always the same value across all scalars) and can be measured with **no superposition loss**. The most informative carries ($H \approx 6$–8 bits) are concentrated in the last 2–3 double-and-add steps, where the accumulator has the most geometric diversity.

Two-register behavior. In the two-register oracle $[a]G + [b]Q$, carries from $[a]G$ and $[b]Q$ collapse the (a, b) superposition independently (Proposition 9). At $f = 90\%$, the two-register equivalence class size is $W_{\text{total}} \approx W_a \times W_b \approx 4$, which is *better* for the QFT than the single-register case. Verified across multiple secret keys ($d = 3, n/2, n-2$) with identical collapse behavior, confirming that carries from the two scalar multiplications are informationally independent.

Remark 10. In circuits with shared sub-computation (e.g., shared adders, windowed precomputation), the carry streams may be correlated. This can only *increase* information leakage per measurement, making collapse *worse* (lower W at the same f). Proposition 9 and the independent-stream model therefore give an *upper bound* on W; the negative result (full measurement destroys superposition) holds *a fortiori* for shared computation.

Concrete resource estimate. Applied to the Chevignard–Fouque–Schrottenloher oracle [3] for secp256k1 ($b = 256$, baseline: 1,193 qubits, ∼60M Toffoli):

At $f = 90\%$: the forward pass costs ∼30M Toffoli; uncomputing 10% of carries costs ∼3M; mid-circuit measurement overhead (∼1,024 $\times$ d syndrome rounds) is negligible. The ∼100–150 qubit savings come from reduced pebble registers.

Scaling law: proof from injectivity. The information rate of 1/16 bits per entry is not merely empirical—it follows from the injectivity results already established.

Table 5: Partial carry-reuse applied to [3].

Configuration	Qubits	Toffoli	Runtime*	Speedup
Baseline [3]	1,193	∼60M	∼60 s	1.00×
+ carry-reuse f=90%	∼1,050	∼33M	∼33 s	1.82×
+ carry-reuse f=85%	∼1,080	∼35M	∼35 s	1.71×
+ carry-reuse f=100%	∼1,000	∼30M	∼30 s	N/A†

*At 1 MHz logical gate rate. †QFT fails; included for comparison only.

Proposition 11 (Information Rate and f^* Scaling)**.** *In MSB-first double-and-add with extended carries, the conditional entropy $H(C_i \mid C_0, \ldots, C_{i-1}) = 1$ bit for each informative step i. Consequently $f^* \geq 1 - 1/b$, and at $b = 256$: $f^* \geq 99.6\%$.*

Proof. **Step 1.** By extended carry injectivity (Theorem 5), the carry prefix $(C_0, \ldots, C_{i-1})$ determines the scalar prefix $(k_{b-1}, \ldots, k_{b-i})$ and hence the accumulator R_i.

Step 2. Given R_i, the step-i carries depend only on the current bit k_{b-1-i}: if $k_{b-1-i} = 0$, only doubling carries are produced (addition carries are all zero); if $k_{b-1-i} = 1$, both doubling and addition carries are produced.

Step 3 (the step-level carries distinguish bit 0 from bit 1). The addition $R_i + G$ computes $x_3 = \lambda^2 - x_{R_i} - g_x$ and $y_3 = \lambda(x_{R_i} - x_3) - y_{R_i}$. At least one subtraction produces a negative intermediate (requiring modular reduction, flag = 1):

- If $\lambda = 0$ (y-matching): Proposition 6 proves a reduction flag differs.
- If $\lambda \neq 0$ and $g_y < y_{R_i}$: the numerator $(g_y - y_{R_i})$ is negative $\Rightarrow$ reduction flag = 1. But in the no-addition case this flag is 0.
- If $\lambda \neq 0$ and $g_y \geq y_{R_i}$: either $g_x < x_{R_i}$ (denominator negative, flag = 1) or $g_x \geq x_{R_i}$ (both non-negative). In the latter case, the multiplication quotient $\lfloor \lambda^2/p \rfloor > 0$ whenever $\lambda \geq \sqrt{p}$, which holds for all but $\mathcal{O}(\sqrt{p})$ values of λ. The residual probability is $< \sqrt{p}/p = p^{-1/2} < 2^{-128}$ at $b = 256$.

Thus: the step-i extended carries distinguish bit 0 from bit 1 with certainty for the y-matching and sign-difference cases (which cover all but a $p^{-1/2}$ fraction), and with probability $> 1 - 2^{-128}$ for the remaining case. Over $b = 256$ steps: Pr[any step fails] $< 256 \cdot 2^{-128} < 2^{-120}$.

Step 4. For a uniformly random scalar, $H(k_{b-1-i}) = 1$ bit. Summing: $H_{\text{total}} = b - 1 \pm 2^{-120}$ bits over $M = 16b$ entries. Rate: $(b-1)/(16b) \approx 1/16$. To preserve $W \geq 2$: measure at most $16(b-1)$ entries, giving $f^* = 1 - 1/b$. □

Validation. Exact conditional-entropy computations at $b = 6$–15 (10 curves, n up to 16,619) confirm: total entropy matches $\log_2 n$ within ± 0.02 bits; per-step conditional entropy converges to 1.00 bit; predicted f^* matches observed within ± 4 pp at every tested b.

At $b = 256$: only ∼16 carries need uncomputation. Overhead: ∼1M gates out of ∼30M = 3.3%. Effective speedup: 60M/31M = **1.94**×.

Caveats. Entries within a step have heterogeneous entropy (33% are zero-entropy), but the per-step total is consistently ∼1 bit (b = 4–15). Entropy-guided ordering (measuring zero-entropy carries first) performs *worse* than prefix ordering because it concentrates informative carries at the end, causing sudden collapse; prefix ordering's gradual interleaving is near-optimal.

6 Discussion

Summary. We characterized carry collisions algebraically, proved extended injectivity across four arithmetic models (affine, Jacobian, Montgomery, fixed-window — 142 curve–model config-

urations), identified the depth–superposition trade-off, and resolved it: partial carry measurement at $f \approx 1 - 1/b$ achieves nearly the full $2\times$ depth reduction while preserving QFT recovery. Applied to the Chevignard et al. oracle [3]: $\sim$31M Toffoli in $\sim$31 s (vs. $\sim$60M baseline), a $\mathbf{1.94}\times$ speedup at $\sim$1,050 qubits.

Scope. The algebraic characterization holds at any scale. Extended injectivity is verified at $b \leq 15$ (affine), $b \leq 13$ (Jacobian), $b \leq 14$ (Montgomery and fixed-window). The scaling law $f^* = 1 - 1/b$ is validated at $b = 4$–14 and predicts $f^* = 99.6\%$ at $b = 256$. Persistence is bounded conservatively ($< 2^{-239}$) using only the first carry per step; per-carry independence would tighten this to $< 2^{-482}$ but changes no practical conclusion. QFT success probability is $\Pr[\text{good}] \approx 0.78$–$0.82 \gg 4/\pi^2$.

Remaining open question. Windowed arithmetic (as distinct from fixed-window, which is tested) uses variable-size windows and more complex scheduling; verifying carry injectivity for the specific windowed decomposition in [3] would confirm the $1.94\times$ speedup for that exact circuit.

Conclusion. Carry-reuse creates a depth–superposition trade-off that vanishes at cryptographic scale. The information rate of 1/16 bits per carry entry (Proposition 11) ensures that at $b = 256$, measuring 99.6% of carries achieves $1.94\times$ depth reduction with QFT success probability $\gg 4/\pi^2$. This optimization is practical for production-grade ECDLP oracles.

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